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Gabriel Rodriguez-Rondon

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Current Position

Senior Economist, Bank of Canada

Since June 2025

Education

Ph.D. in Economics, McGill University 2025
Thesis: “Monte Carlo Test Methods for Markov Switching and Stochastic Volatility Models with Applications in Macroeconomics and Finance”
Committee: Jean-Marie Dufour (Chair), Zhongjun Qu, John Galbraith, & Hamed Bouakez
Visiting Ph.D. Student, Boston University Spring 2023
M.A. in Economics, McGill University 2019
B.A. (Honours) in Economics, Carleton University 2016

Research Interests

Econometrics, Macroeconomics and Monetary Economics, Finance, Machine Learning

Refereed Publications

1. “MSTest: An R Package for Testing Markov Switching Models”, with Jean-Marie Dufour, *The R Journal*, vol. 18, no. 3: 217–239. doi: 10.32614/RJ-2026-044.
2. “Estimation and Inference for Higher-Order Stochastic Volatility Models With Leverage”, with Jean-Marie Dufour & Md. Nazmul Ahsan, *Journal of Time Series Analysis*, vol. 46, no. 6: 1064–1084. doi: 10.1111/jtsa.12851.

Conference Proceedings

3. “Simulation-Based Inference for the Synchronization of Business Cycles”, with Jean-Marie Dufour, *Proceedings of the Business and Economic Statistics Section of the American Statistical Association*, December 2023.
4. “Simulation-Based Inference for Markov Switching Models”, with Jean-Marie Dufour, *Proceedings of the Business and Economic Statistics Section of the American Statistical Association*, 275–289, December 2022.

Working Papers

5. “Monte Carlo Likelihood Ratio Tests for Markov Switching Models”, with Jean-Marie Dufour, *Bank of Canada Staff Working Paper*, no. 2026-23. doi: 10.34989/swp-2026-23. Revised February 2026.
6. “Underlying Core Inflation with Multiple Regimes”, Discussion Paper, McGill University. arxiv.org/abs/2411.12845. Revised November 2024.

7. “Implied Factors: The Linear Skeleton of Machine Learning Forecasts”, with Philippe Goulet Coulombe & Karin Klieber. Revised June 2026.
8. “wARMASVp: Estimation, Inference, Filtering, and Forecasting for Higher-Order Stochastic Volatility Models in R”, with Jean-Marie Dufour & Md. Nazmul Ahsan. Revised May 2026.
9. “Volatility Forecasting with Higher-order Stochastic Volatility Models”, with Jean-Marie Dufour & Md. Nazmul Ahsan, Discussion Paper, McGill University. Revised June 2026.
10. “Estimation and inference for stochastic volatility models with leverage and heavy-tailed distributions”, with Jean-Marie Dufour & Md. Nazmul Ahsan, Discussion Paper, McGill University, doi: 10.34989/swp-2026-8. Revised June 2026.
11. “Joint Determination of Counterparty and Liquidity Risk in Payment Systems”, with Jorge Cruz Lopez & Charles M. Kahn. Revised September 2024. Awarded *Best Paper on Risk Management* at the NFA 2019 Conference.

Policy & Technical Notes

12. “Practical estimation of high-dimensional stochastic volatility models with application to macroeconomic uncertainty in Québec and Canada”, with Jean-Marie Dufour & Md. Nazmul Ahsan, 2025RP-19, Projects Reports, CIRANO. doi: 10.54932/XBOU8314
13. “The Government of Canada Debt Securities Dataset”, with Jeffrey Gao & Francisco Rivadeneyra, *Bank of Canada Technical Report*, February 2018.

Work in Progress

14. “Practical and reliable estimation methods for high-dimensional multivariate stochastic volatility models with macroeconomic applications”, with Jean-Marie Dufour & Md. Nazmul Ahsan.
15. “Monte Carlo Test for Factor Models with Markov switching”
16. “MNbreaks: An R Package for Estimating and Testing Multiple Structural Changes in Multivariate Linear Regression Models”, with Pierre Perron & Zhongjun Qu.

Grants & Awards

Fonds de Recherche du Québec - Société et Culture (FRQSC), PhD Fellowship	2021 - 2024
Graduate Excellence Award, McGill University	2021 - 2024
International Association for Applied Econometrics Travel Grant	2023
McGill Graduate Mobility Award	2022 - 2023
McGill Graduate Research Enhancement and Travel Awards	2022 - 2023
Canadian Economics Association Travel Grant	2022
Carleton Academic Scholarship	2012 - 2013

Conference & Seminar Presentations

“Implied Factors: The Linear Skeleton of Machine Learning Forecasts” (with Philippe Goulet Coulombe & Karin Klieber)

- BSE Summer Forum Workshop on AI and Machine Learning in Economics, Barcelona, Spain, June 11, 2026.
- ECONDAT 2026 Spring Meeting, Bank of France, Paris, France, June 5, 2026.

- 60th Annual Meetings of the Canadian Economics Association, Vancouver, British Columbia, May 29, 2026.
- Bank of Canada Fellowship Learning Exchange, Ottawa, Ontario, April 30, 2026.

“Monte Carlo Likelihood Ratio Tests for Markov Switching Models” (with Jean-Marie Dufour),

- The Econometric Society, World Congress, Seoul, Korea, August 18, 2025.
- CIREQ Econometrics Conference: Recent Developments in Identification and Inference, Montreal, Quebec, April 18, 2025.
- Universidad del Rosario (Virtual) Seminar, Bogota, Colombia, April 8, 2025.
- CIREQ-McGill Lunch Seminar, Montreal, Quebec, October 8, 2024.
- 76th European meeting of the Econometric Society, Erasmus School of Economics, Rotterdam, Netherlands, August 26, 2024.
- New York Camp Econometrics XVIII, Lake Placid, NY, April 26, 2024.
- Carleton University Brown Bag Seminar, Ottawa, Ontario, October 19, 2023.
- NBER-NSF Time Series Conference, Department of Economics, Université du Québec à Montréal, Montreal, Quebec, September 23, 2023.
- International Association for Applied Econometrics (IAAE) Annual Conference, BI Norwegian Business School, Oslo, Norway, June 28, 2023.
- Boston University Econometrics Seminar, Boston, MA, February 24, 2023.
- 16th International Conference on Computational and Financial Econometrics, King’s College, London, UK, December 18, 2022.
- Latin American Meeting of the Econometric Society (LAMES), Universidad del Pacífico, Lima, Peru, November 3, 2022.
- Joint Statistical Meetings, American Statistical Association, Washington, D.C. August 9, 2022.
- 17th CIREQ Ph.D. Students’ Conference, Concordia University, Montreal, Quebec, June 6, 2022.
- 56th Annual Meetings of the Canadian Economics Association, Carleton University, Ottawa, Ontario, June 3, 2022.

“Volatility Forecasting with Higher-order Stochastic Volatility Models”

- International Association for Applied Econometrics (IAAE) Annual Conference, Torino, Italy June 25, 2025.

“Underlying Core Inflation with Multiple Regimes”

- New York Camp Econometrics XIX, Bolton Landing, NY, April 25, 2025.
- Bank of Canada Conference on Real-Time Data Analysis, Methods and Applications in Macroeconomics and Finance, Ottawa, Ontario, October 17, 2024.
- International Association for Applied Econometrics (IAAE) Annual Conference, Thessaloniki, Greece June 27, 2024.
- Bank of Canada Brown Bag Seminar, Ottawa, Ontario, July 5, 2023.
- 57th Annual Meetings of the Canadian Economics Association, Winnipeg, Manitoba, June 3, 2023.

“Estimation and Inference for Higher-order Stochastic Volatility Models with Leverage” (with Jean-Marie Dufour & Md. Nazmul Ahsan)

- 39th Annual Meeting of the Canadian Econometrics Study Group (CESG), Toronto, Ontario, October 25, 2024.

- North American Summer Meeting of the Econometric Society, Vanderbilt University, Nashville, TN, June 13, 2024.
- 58th Annual Meetings of the Canadian Economics Association, Toronto Metropolitan University, Toronto, Ontario, May 31, 2024.
- 63e Congrès de la Société Canadienne de Science Économique, Montreal, Quebec, May 15, 2024.
- CIREQ Econometrics Conference in Honor of Eric Ghysels, Montreal, Quebec, May 10, 2024.

“Joint Determination of Counterparty and Liquidity Risk in Payment Systems” (with Jorge Cruz Lopez & Charles M. Kahn),

- II Regional Conference on Payments and Financial Market Infrastructures, Banco de la República de Colombia & CEMLA, Bogota, Colombia, September 21, 2023.
- Payments Canada & Bank of Canada Research Symposium, Ottawa, Ontario, October 2017.
- 51st Annual Meetings of the Canadian Economics Association, St. Francis Xavier University, Antigonish, Nova Scotia, June 3, 2017.

“Testing for the Synchronization of International Business Cycles” (with Jean-Marie Dufour)

- Joint Statistical Meetings, American Statistical Association, Toronto, Ontario, August 10, 2023.

Teaching Experience

Lecturer, HEC Montréal - Time Series Econometrics (MSc.)	Winter 2026, 2027
Lecturer, HEC Montréal - Financial Econometrics (MSc.)	Winter 2025
Guest Lecturer, McGill University: Econometrics II (Ph.D.)	Winter 2024
Teaching Assistant, HEC Montréal	2024 - 2025
Teaching Assistant, McGill University	2018 - 2025

Research Experience

Research Assistant, McGill University - Prof. Jean-Marie Dufour	2019 - 2025
Research Assistant, UQAM - Prof. Philippe Goulet Coulombe	2024 - 2025
Ph.D. Intern, Bank of Canada	2023 - 2024
Research Assistant, McGill University - Prof. Francesca Carrieri	2023 - 2024
Research Data Scientist, Financial Network Analytics	2019 - 2022
Research Assistant, Bank of Canada	2016 - 2018

Professional Service

Referee

Economic Modelling
 International Statistical Review
 Journal of Financial Market Infrastructures
 Journal of International Financial Markets, Institutions & Money
 Journal of Portfolio Management
 Journal of Quantitative Economics

Discussant

CEMLA - II Conference on Payments and FMIs (2023)
 58th Annual Meetings of the Canadian Economics Association (2024)

Conference/Session Organization

Bank of Canada Special Session on “Advances in High-Dimensional Time Series and Macroeconomic Forecasting”, Canadian Economics Association Annual Conference (2026)

Outreach & Service

CEA-Bank of Canada Undergraduate Poster Competition (2026)

Governor’s Challenge Steering Committee, Chair (2026-2027)

Governor’s Challenge Steering Committee (2025-2026)

Computational Skills

R, Python, MATLAB, Dynare, C++, Linux, GAUSS, STATA

Software

wARMASVp - R package, creator & maintainer, available on CRAN

MSTest - R package, creator & maintainer, available on CRAN

MaxMC - R package, maintainer, available on CRAN

mbreaks (v2) - R package, creator & maintainer, available on GitHub

bootBtest - R package, creator & maintainer, available on GitHub

Languages

English (native), French (fluent), Spanish (fluent)

Citizenship

Canada, Peru

References

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